

Certified Position–Topology Trade-offs for Hierarchy-Constrained 1-D Layouts

Category: Research

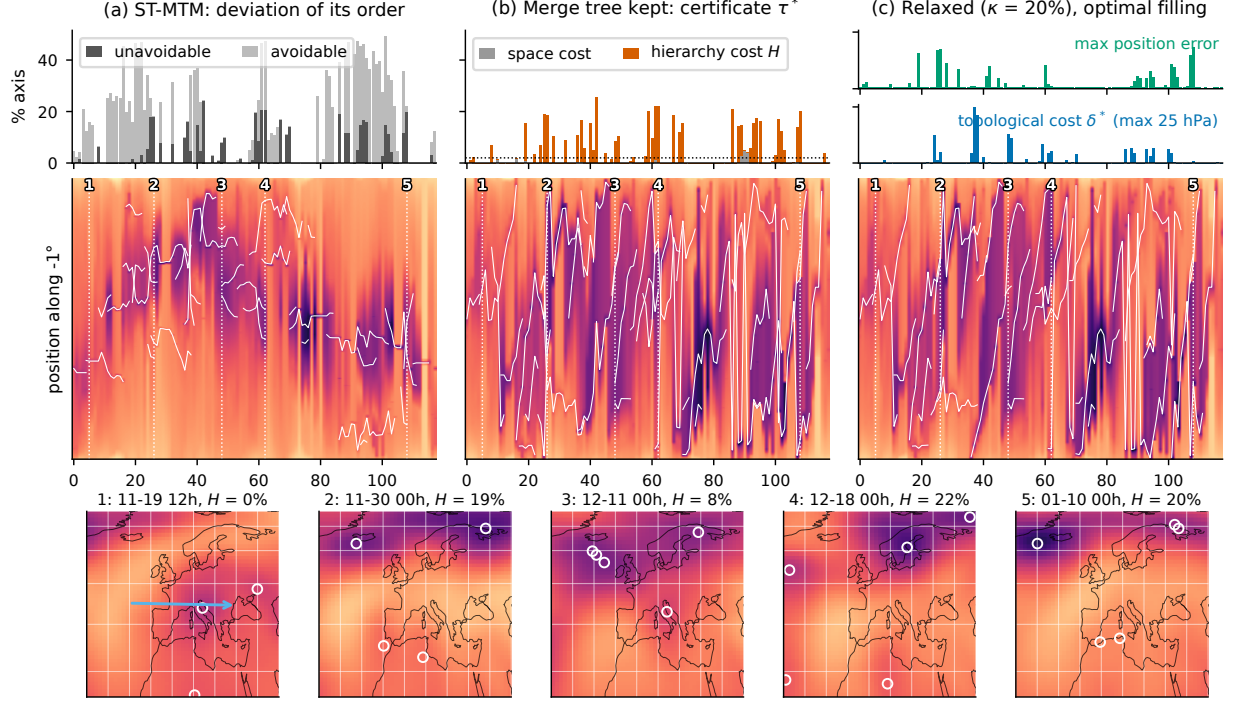


Figure 1: ERA5 pressure lows over 59 days. (a) ST-MTM: unavoidable and avoidable order error after calibration. (b) Tree-preserving anchored map: certified space and hierarchy costs. (c) Relaxed map: actual position error and topological cost in hPa. The strips measure different quantities; lower panels show the pressure fields.

ABSTRACT

Hierarchy-constrained 1-D displays keep related features contiguous, but this order can conflict with a spatial reference. We characterize the resulting position–topology trade-off for merge tree maps. A positional certificate gives the least possible maximum reference error, separates hierarchy cost from packing cost, and identifies conflicting features. We derive a fixed-order closed form, an exact dynamic program at display resolution for bounded-degree hierarchies, and a width-free bound for existing maps. Breaking subtree contiguity incurs a filling-independent merge-level error; an optimal value-preserving filling then yields the exact trade-off frontier for small trees. These results support a diagnostic and relaxation method that reports both position error and topological cost. On two ERA5 periods and an Australian wildfire sequence, the chosen extraction protocols produce hierarchy conflicts in 24–63% of frames. Relaxation reduces geometric motion-proxy reversals in several settings, at substantial and data-dependent topological cost; it does not uniformly improve neighbourhood preservation or establish a benefit to human readers. A 1024-row dendrogram example demonstrates the positional certificate beyond merge tree maps. The complete relaxation and exact-frontier computations remain limited by order enumeration and node degree.

Index Terms: Merge trees, hierarchy-constrained layout, time-varying scalar fields, static visualization, faithfulness, interleaving distance.

1 INTRODUCTION

A static summary shows the complete evolution of spatio-temporal data in a single image. It complements animation, which is transient and makes comparisons across time hard. Hovmöller diagrams track weather systems along a fixed longitude or latitude [18]; dense pixel displays and MotionRugs arrange entities as pixel columns over time [20, 4]; 1-D projections support the analysis of spatio-temporal phenomena [15]; and merge tree maps linearize each time step of a scalar field while preserving its merge tree [24, 8]. All of these designs compress space to one axis and put time on the other. Readers then infer spatial proximity and motion from vertical proximity: two features drawn close together are read as close, and two tracks that converge are read as approaching.

These displays share a second property: *their 1-D order is constrained by a hierarchy*. Merge tree maps keep the leaves of every merge-tree subtree contiguous, which keeps features intact and makes the merge tree readable from the image [24]. Clustered heatmaps [12] and hierarchical orderings of pixel displays [15] keep clusters contiguous, and geophylogenies align tree-constrained leaf orders with map sites [22]. Hierarchy and space need not agree. Two pressure lows may merge first because they share a trough, while an unrelated third low lies geographically between them (Fig. 2). Every admissible layout must then misplace at least one of the three, and the reader of the map has no way of knowing which one, or by how much.

Abstract views are known to mislead [29, 45], and projection quality is routinely measured [44, 1]. ST-MTM reports distance

stress and neighbourhood preservation and notes that no 1-D order preserves all distances [8]. Such measures describe the layout that was produced. No existing method certifies the *unavoidable* deviation from a positional reference, identifies which features cause it, or states how much hierarchy information must be given up to reduce it. The consequence is concrete: in a geometric reader proxy on ERA5 sea-level pressure (Sec. 7.3), the vertical distance change of two lows on a temporal merge tree map has the opposite sign of their true 24-hour approach or separation in 26.9% of clear cases, and 21.2% on ST-MTM (permutation null: 35–36%), whereas a map that draws every low exactly at its reference position errs in 10.6%.

We therefore ask: **how faithful to a positional reference can a hierarchy-constrained 1-D layout be, and can its unfaithfulness be certified, located, and controlled?** Following the argument that visualization quality must be defined before it can be measured and guaranteed [30], we do not propose yet another layout heuristic. Instead we characterize what *any* layout can achieve, and then use this characterization to build, disclose, and relax layouts. Our contributions are:

- A **positional certificate** for hierarchy-constrained interval layouts, including fixed-order and width-free forms, witness triples, and an exact display-resolution algorithm for bounded-degree trees (Sec. 4).
- A **position–topology trade-off**: a filling-independent lower bound, an optimal value-preserving filling, and an exact frontier for small trees (Sec. 5).
- A **diagnostic and relaxation method**, evaluated on real and synthetic fields with explicit accounts of projection loss, preprocessing sensitivity, and topological cost (Secs. 6 and 7).

2 RELATED WORK

Static summaries of spatio-temporal data. Hovmöller and radius–time diagrams slice or aggregate one dimension; space-filling curves and pixel techniques linearize space [15, 51, 20]. MotionRugs, stable visual summaries, MoReVis, and GroupRugs summarize trajectories and regions [4, 48, 42, 40]. GroupRugs evaluates neighbourhood-based spatial quality under group contiguity but does not bound unavoidable conflict. Topology-based summaries include topological landscapes [46, 3], temporal merge tree maps (TMTM) [24], spatiotemporal merge tree maps (ST-MTM) [8], temporal treemaps and nested tracking graphs [23, 28]. TMTM favors overlap between columns; ST-MTM additionally fits geometric distances and widths. Both preserve the tree but do not certify positional deviation.

Hierarchy-constrained orders. Optimal leaf ordering [2] underlies clustered heatmaps [12]. Tree-constrained leaf orders against external positions appear in geophylogenies [22], in re-ordering a tree according to a given leaf order [5], and in displacement-optimized tanglegrams [19]; the objectives there are crossings, inversions, or summed displacement. Storyline layouts such as StoryFlow also order lines under hierarchy constraints and minimize crossings and wiggles [27]. We address a bottleneck objective (the maximum deviation), include widths, gaps, a canvas, and anchor eccentricity, and pair it with a topological cost.

Faithfulness, distortion, and trade-offs. Visualization mirages and misread embeddings motivate explicit distortion analysis [29, 45]. Trustworthiness, continuity, and distortion maps assess projections [44, 1, 38, 37]. Cartograms, data–spatial grid layouts, and spatially ordered treemaps trade or measure geometric fidelity [35, 43, 47]. Formal accounts of data–visual correspondence and task-specific projection distortion [21, 33, 34] assess produced views. We instead bound distortion over all admissible layouts and compute an attainable trade-off for small trees.

1-D placement. Without a hierarchy, minimizing the maximum displacement needed to separate intervals on a line is solvable in $O(n \log n)$ [26]; this is the free case of our certificate without canvas and eccentricity constraints. Separation-constrained quadratic programs are standard in constrained graph layout [10], and we use one to choose a layout within the certified budget.

Topological simplification. Persistence-based simplification removes low-persistence extrema and thereby simplifies merge trees [11, 6, 41]. It could also remove conflicts, but only by deleting features. Our flattening keeps every feature and removes a merge relation; its cost γ_i is the gap between two adjacent merge values, not the persistence of an extremum. Section 7.2 compares the two.

Merge tree distances. Interleaving distances compare merge trees [31]; the ℓ_∞ -cophenetic metric of labelled trees is an interleaving distance [32]. Intrinsic and geometry-aware interleavings and Wasserstein metrics support comparison, tracking, and averaging of time-varying data [16, 50, 36], including tropical cyclones and atmospheric rivers [49, 25]. Persistent homology has been used to evaluate and construct projections [39, 9], and the merge distortion metric compares cluster trees [13]. We use the labelled interleaving distance to measure what a 1-D map does to the specific merge tree it depicts.

3 PROBLEM AND MOTIVATION

3.1 Merge tree maps

For each time step t of a scalar field f_t on a 2-D domain, the merge tree T_t records how the components of sublevel sets (a *join tree*, e.g., for pressure minima) or superlevel sets (a *split tree*, e.g., for fire-intensity maxima) are born at extrema and merge at saddles. Its leaves are the features of interest. Leaf i has an extremum position $R_i \in \mathbb{R}^2$ and a support measure A_i (the area of its segment), and every pair of leaves has a merge value $f(\text{lca}(i, j))$. A merge tree map chooses, per time step, a leaf order, an interval per leaf, and an anchor inside the interval; fills the column by sampling values along leaf arcs and lowest-common-ancestor (LCA) paths [24, 8]; and stacks the columns over time. Because every subtree occupies a contiguous block, the merge levels read from the column equal those of T_t : the map preserves topology.

3.2 Tasks and the role of the reference

We consider three tasks that readers perform on a merge tree map. (T1) *Identify and compare features*: which extrema exist, how deep they are, and how they merge. (T2) *Locate*: where a feature lies relative to others along a spatial direction. (T3) *Follow motion*: whether two features approach or separate over time. T1 needs topology and size; T2 and T3 need position. Merge tree maps were designed for T1 and support it exactly. T2 and T3 are nevertheless performed on them: vertical proximity is the only spatial cue the image offers, ST-MTM places leaves to reflect geometric distances [8], and 1-D projections are used to follow spatial phenomena [15].

What the reference is, and is not. We make the positional reading explicit through a *reference coordinate* $q_i = \phi(R_i)$, the 1-D coordinate of feature i that a reader should be able to recover from its vertical position. The reference is a property of the task, not of a method, and all our results hold for any given reference. In the evaluation we use the principal direction of feature motion, which serves T3, and report fixed directions for robustness. The certificate therefore does not claim that TMTM or ST-MTM are wrong; they optimize other goals. It states what *any* merge-tree-preserving map, theirs included, can offer a reader who performs T2 or T3 with that reference.

Why the maximum deviation. We bound the maximum rather than an average deviation. A reader does not know which feature is misplaced, so a guarantee must hold for every feature. Averages

and stress hide the few misplaced features on which motion readings reverse (Sec. 7.3). And the maximum admits a closed form, an exact algorithm, and a clean interaction with topology (Sec. 5). Average-type measures remain useful for choosing among layouts within the certified budget, which is how we use them (Sec. 6).

Why optimize the bound? A reference-anchored layout minimizes maximum deviation under the hierarchy, then balances secondary quality criteria within a small slack. On ERA5 it still exceeds 2% of the axis in 34% of frames and reverses 19.2% of clear geometric motion-proxy readings, compared with 21.2% for ST-MTM. This motivates separating unavoidable hierarchy cost from layout choices, rather than attributing all error to an optimizer.

3.3 Model

Let the hierarchy be a rooted tree T with leaves $1, \dots, n$. Leaf i has a reference q_i , a width $w_i = cA_i$ with a scale c that is fixed over time, and an eccentricity radius $h_i = \rho w_i/2$ with $\rho \in [0, 1]$. Let $g > 0$ be a minimum gap and $[a, a + L]$ the canvas. A layout (π, z, u) consists of a leaf order π , interval centres z , and anchors u that satisfy

- (H) $\pi \in \Pi(T)$: the leaf set S_v of every node v is contiguous in π ;
- (D) $z_y - z_x \geq (w_x + w_y)/2 + g$ for π -consecutive leaves x, y ;
- (C) $a + w_i/2 \leq z_i \leq a + L - w_i/2$;
- (E) $|u_i - z_i| \leq h_i$.

Legal orders arise by permuting the blocks of children at every node, so $|\Pi(T)| = \prod_v \deg(v)!$. We distinguish three fidelities: *position*, the deviation $u_i - q_i$; *size*, widths proportional to support measure and comparable over time; and *topology*, the merge levels shown by the rendered map (Sec. 5). TMTM preserves topology and absolute size but does not control position; ST-MTM preserves topology, reflects relative distances, and fits target width shares; the reference-anchored layout targets position under (H). Our analysis concerns position and topology at fixed size.

4 A POSITIONAL CERTIFICATE

4.1 Closed form and decomposition

Definition 1. For an order π , $\tau(\pi) = \min_{z,u} \max_i |u_i - q_i|$ subject to (D), (C), (E). The certificate is $\tau^*(T) = \min_{\pi \in \Pi(T)} \tau(\pi)$, and τ_{free} is the same minimum over all $n!$ permutations. The hierarchy cost is $H = \tau^* - \tau_{\text{free}}$ and the space cost is τ_{free} .

τ^* is a property of the data and the hierarchy, not of an algorithm: no layout that keeps every subtree contiguous can deviate by less, and some layout attains it. For a fixed order it can be written down directly.

Proposition 2 (Closed form). *Number the leaves along π and let $s_k = (w_k + w_{k+1})/2 + g$, $S_{ij} = \sum_{k=i}^{j-1} s_k$, $D_k = \sum_{m < k} w_m + w_k/2 + (k-1)g$, and $E_k = \sum_{m > k} w_m + w_k/2 + (n-k)g$. If $\sum_i w_i + (n-1)g \leq L$, then*

$$\tau(\pi) = \max \left\{ 0, \max_{i < j} \frac{1}{2} (q_i - q_j + S_{ij} - h_i - h_j), \max_k (a + D_k - q_k - h_k), \max_k (q_k - h_k - a - L + E_k) \right\}.$$

Eliminating the anchors turns each constraint into a box window $|z_k - q_k| \leq \tau + h_k$ for the centre. A chain with lower-bounded separations and box windows is feasible if and only if its earliest placement stays inside every window, and expanding the resulting inequalities gives the formula. The pair term says that two leaves in inverted reference order must share the overlap of their windows; the other two terms account for the canvas ends. The formula agrees with a linear-programming solver to $2 \cdot 10^{-14}$ on our data.

Algorithm 1 Certificate τ^* at display resolution (Theorem 4).

Require: hierarchy T ; pixel widths W_i , gap G , references q_i , radii h_i ; N pixels; tolerance ϵ

- 1: **function** EARLIEST(S, τ) ▷ array over starts $x = 0, \dots, N$
- 2: **if** S is a leaf i **then**
- 3: $lo_i \leftarrow \max(0, \lceil q_i - \tau - h_i - W_i/2 \rceil)$, $hi_i \leftarrow \min(N - W_i, \lfloor q_i + \tau + h_i - W_i/2 \rfloor)$
- 4: **return** $x \mapsto \max(x, lo_i) + W_i + G$ if $\max(x, lo_i) \leq hi_i$,
 else ∞
- 5: $E_c \leftarrow \text{EARLIEST}(c, \tau)$ for every child c of S
- 6: $F_\emptyset \leftarrow \text{id}$; $F_M \leftarrow \min_{c \in M} E_c \circ F_{M \setminus \{c\}}$ for all subsets M of children
- 7: **return** $F_{\text{all children}}$ ▷ binary: $\min(E_B \circ E_A, E_A \circ E_B)$
- 8: $\tau^* \leftarrow$ smallest τ with $\text{EARLIEST}(\text{root}, \tau)(0) < \infty$, by bisection to ϵ

Proposition 3 (Monotonicity, decomposition, witnesses). *Flattening a node, i.e., attaching its children to its parent, enlarges Π and never increases τ^* ; flattening all nodes yields τ_{free} . Hence $H \geq 0$ is exactly the price of keeping every subtree contiguous. If $i, j \in S_v$, $k \notin S_v$, and $q_i < q_k < q_j$, then $\tau^* \geq \frac{1}{2} \min(q_k - q_i, q_j - q_k)$.*

The last statement holds because contiguity of S_v forces u_k outside $[u_i, u_j]$, so one of the three anchors must cross a reference that lies between the others. Such a *witness triple* names the conflicting features. In Fig. 2, $(A, B; C)$ gives the bound 1.4, and the widths raise τ^* to 1.85. On the real datasets, the best witness explains a median of 92–94% of τ^* in conflict frames, so a conflict is usually caused by one identifiable triple of features.

4.2 Exact computation at display resolution

Enumerating $\Pi(T)$ is exponential in the number of internal nodes. We therefore compute τ^* at the resolution of the display. Discretize the canvas into N pixels with integer widths W_i and gap G , and let $E_S(x)$ be the earliest start that is available after placing all leaves of subtree S in some legal order, none starting before x .

Theorem 4 (Exact dynamic program). *For a leaf, $E_i(x) = \max(x, lo_i) + W_i + G$, or ∞ if $\max(x, lo_i) > hi_i$, where $[lo_i, hi_i]$ is the window of Proposition 2. For a binary node, $E_S = \min(E_B \circ E_A, E_A \circ E_B)$; for an m -ary node, a subset recursion over its children. A deviation τ is feasible iff $E_{\text{root}}(0) < \infty$. Bisection yields τ^* to a tolerance ϵ in $O(nN \log(N/\epsilon))$ time for binary hierarchies—pseudo-polynomial in N —and $O(2^m mN)$ per m -ary node.*

Correctness rests on an earliest-completion argument. For a fixed order, the greedy earliest placement minimizes every start; hence a block influences the rest of the chain only through its earliest next start, and a smaller next start is never worse. The functions E_S are monotone arrays over $N+1$ start positions, and composing them is an array lookup. Algorithm 1 lists the computation.

Proposition 5 (Certified lower bound). *If widths and gap are integer multiples of the pixel size λ and the canvas is grid-aligned, then $0 \leq \tau_{\text{disc}}^* - \tau_{\text{cont}}^* \leq \lambda/2$. For real widths, rounding widths and gap down while keeping the original slacks gives $\tau_{\text{cont}}^* \geq \tau_{\text{disc, relax}}^* - \lambda/2$.*

Every continuous layout remains feasible for the relaxed discrete problem after shifting its starts to the pixel grid, which costs at most half a pixel; reported values also subtract the bisection tolerance. All values of the lower frontier Φ reported in Sec. 7 use this certified bound. On 300 random hierarchies the program matches brute-force enumeration within 0.5 pixels; on all ERA5 and Ring frames, whose widths are not integer pixels, within 0.6 pixels at about 2 ms per frame; and a 4096-leaf hierarchy on 16384 pixels

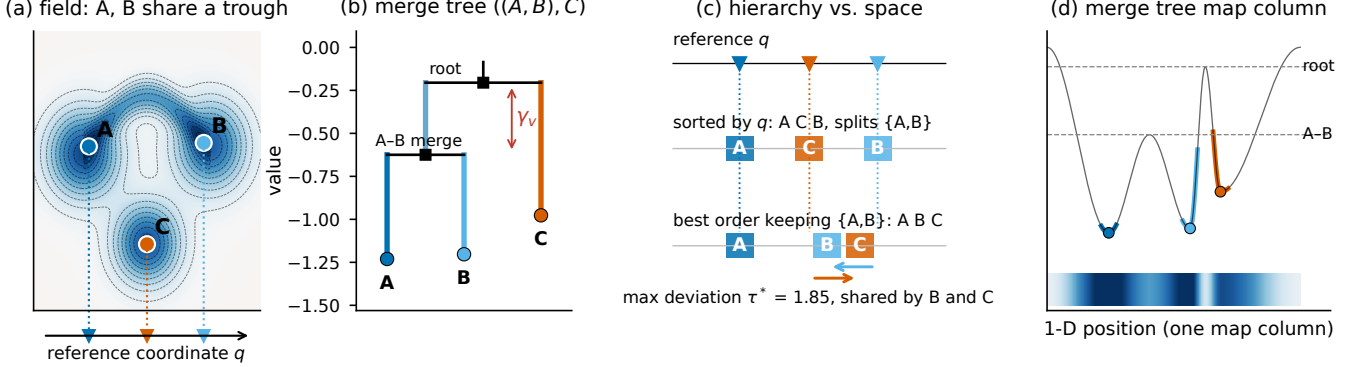


Figure 2: Why hierarchy and space conflict. (a) A scalar field with three minima; A and B share a trough, and C lies between them along the reference direction (dotted: projections q). (b) Its join tree $((A,B),C)$, computed from the field; γ_v is the gap between the A-B merge and the root. (c) Sorting by the reference splits the subtree $\{A,B\}$; the best layout that keeps it contiguous must deviate by τ^* , shared by B and C (arrows). (d) The resulting column of a merge tree map: leaf values at the anchors and the merge values between them, so that the column shows the merge tree.

takes 3.6 s. The scalable part is the certificate; choosing a layout within the budget still enumerates candidate orders (Sec. 6).

4.3 A width-free certificate

Other merge tree maps use other width models: TMTM allocates one pixel per sample, and ST-MTM fits target widths. To bound them in their own models, we drop widths, gaps, eccentricity, and canvas, which removes every constraint except the order.

Proposition 6 (Width-free certificate). *Let $\tau_{pt}(\pi) = \max\{0, \max_i \text{before } j (q_i - q_j)/2\}$ and $\tau_{pt}^* = \min_{\pi \in \Pi(T)} \tau_{pt}(\pi) \leq \tau^*$. For every 1-D map whose anchor order is hierarchy-consistent, with any width model, and for every non-decreasing calibration c of its axis onto the reference axis, $\max_i |c(u_i) - q_i| \geq \tau_{pt}^*$.*

Two anchors in inverted order cannot both be closer than half their reference gap, whatever the calibration, and $\tau_{pt}(\pi)$ is attained by $c(u_k) = \frac{1}{2}(\max_{i < k} q_i + \min_{j > k} q_j)$. The width-free certificate therefore separates, for any existing map, the deviation that its hierarchy forces from the deviation that its layout chose (Fig. 1a).

5 THE PRICE OF TOPOLOGY

5.1 Measuring topology on a 1-D map

Relaxing (H) can reduce the positional error, but the map then no longer shows the merge tree exactly. To quantify the change, consider any function ψ on the line (a *filling*) with leaf anchors u_i . In a join tree, the 1-D merge level of leaves i, j is $m_\psi(i, j) = \max_{[u_i, u_j]} \psi$, because the components of sublevel sets on a line are intervals (split trees are symmetric). We define the *topological distortion*, the largest error of an anchor-to-anchor merge level,

$$d_{\text{top}} = \max_{i \neq j} |m_\psi(i, j) - f(\text{lca}_T(i, j))|.$$

When the local minima of the column are exactly the labelled anchors, with matching leaf values, d_{top} is the ℓ_∞ -cophenetic distance between T and the merge tree of the column, i.e., their labelled interleaving distance [32]. It is zero exactly when the column shows the merge tree. Throughout, critical values are distinct (so $\gamma_v > 0$), statements are given for join trees, and split trees follow by negating f .

5.2 A lower bound for every filling

Theorem 7 (Topological price of non-contiguity). *If the leaf set of a non-root node v is not contiguous in a 1-D map, then for every filling ψ , $d_{\text{top}} \geq \gamma_v/2$ with $\gamma_v = f(p(v)) - f(v)$, where $p(v)$ is the parent of v . The factor $1/2$ is tight.*

Proof. Choose $i, j \in S_v$ and $k \notin S_v$ with u_k between u_i and u_j . Interval inclusion gives $m_\psi(i, k) \leq m_\psi(i, j)$, whereas $m_T(i, j) \leq f(v)$ and $m_T(i, k) \geq f(p(v))$. Hence $\gamma_v \leq m_T(i, k) - m_T(i, j) = [m_T(i, k) - m_\psi(i, k)] + [m_\psi(i, k) - m_\psi(i, j)] + [m_\psi(i, j) - m_T(i, j)] \leq 2d_{\text{top}}$. For tightness, take $T = ((i, j)_v, k)$ with $f(k) \leq f(v) + \gamma_v/2$, anchors in order i, k, j , and both barriers at $f(v) + \gamma_v/2$. $\square$

Figure 3 illustrates the theorem and the optimal filling below on the example of Fig. 2. The theorem constrains the map, not the algorithm: it holds for TMTM, ST-MTM, and any future method that breaks contiguity. Weak merges (small γ_v) are cheap to break; strong merges are expensive.

5.3 The optimal filling

Theorem 7 bounds every filling from below. For a fixed order, the best *value-preserving* filling, i.e., one that shows every extremum at its true value, can be given in closed form. Number the leaves along the map, keep the true leaf values at the anchors, and let $P_{ij} = f(\text{lca}(i, j))$ for $i < j$, $C_k = \min_{i \leq k < j} P_{ij}$, and $\ell_k = \max(f(k), f(k+1))$.

Proposition 8 (Optimal filling). *Over value-preserving fillings, $\inf_\psi d_{\text{top}} = \delta^*(\pi) = \max\{0, \max_k (\ell_k - C_k), \max_{i < j} \frac{1}{2}(P_{ij} - \max_{i \leq k < j} C_k)\}$. It is attained by the barrier filling, whose maximum between anchors k and $k+1$ is $b_k = C_k + \delta^*(\pi)$; if every anchor must remain a strict minimum, it is approached arbitrarily closely by raising barriers with $b_k = \ell_k$ by any $\varepsilon > 0$. Moreover, $\delta^*(\pi) = 0$ iff $\pi \in \Pi(T)$.*

Any filling acts on merge levels only through its barrier heights b_k between consecutive anchors, with $m_\psi(i, j) = \max_{i \leq k < j} b_k$. The upper constraints $b_k \leq P_{ij} + \delta$ collapse to $b_k \leq C_k + \delta$, and raising a barrier only helps the lower constraints, so $b_k = C_k + \delta$ is optimal; feasibility then reduces to the two inequalities in the formula. The term $\ell_k - C_k$ reveals a second mechanism next to Theorem 7: a shallow extremum placed between two leaves that merge deeply raises their 1-D merge level at least to its own value, whatever value-preserving filling is used. If a filling may also move

extremum values, which the labelled interleaving distance permits, this term halves to $\frac{1}{2}(\ell_k - C_k)$; we keep extrema at their true values because readers read depths from them. The pair term has the structure of ℓ_∞ fitting by ultrametrics [14, 7], here constrained to the ultrametrics that a line with fixed leaf values can realize. The LCA-path filling of existing maps is not optimal; on our data it pays up to twice δ^* . The barrier filling changes a rendered column only between anchors, by clipping each segment to b_k and raising its existing peak to b_k if necessary, so no new extrema appear; our renderer raises barriers by about 10^{-6} relative to the field values above adjacent leaf values, so every anchor stays a strict extremum and every feature remains visible.

5.4 The position–topology frontier

Corollary 9 (Certified lower frontier). *Let $\Phi_T(\delta)$ be τ^* of T with every node of $\gamma_v \leq 2\delta$ flattened. Every layout satisfying (D), (C), (E), with any filling, obeys $\max_i |u_i - q_i| \geq \Phi_T(d_{\text{top}})$.*

By Theorem 7, every non-contiguous node of a layout with distortion d_{top} has $\gamma_v \leq 2d_{\text{top}}$, so the layout is legal for the flattened tree, whose certificate is $\Phi_T(d_{\text{top}})$ (Proposition 3). No 1-D map can simultaneously keep the positional error below $\Phi_T(\delta)$ and the topological distortion at or below δ . Φ_T is a step function with breakpoints at $\gamma_v/2$, and we evaluate it per frame with the certified lower bound of Proposition 5. It is generally not attainable. Computing it with the DP remains exponential in the largest child degree after flattening, which can be much larger than in the input tree. For small trees, the attainable frontier itself can be computed exactly.

Theorem 10 (Exact position–topology frontier). *A value-preserving map with order π has $\max_i |u_i - q_i| \geq \tau(\pi)$ and $d_{\text{top}} \geq \delta^*(\pi)$, and both bounds are attained, or approached arbitrarily closely, simultaneously. Hence the achievable pairs over all value-preserving 1-D maps have the lower envelope $F_T(\delta) = \min\{\tau(\pi) : \pi \in S_n, \delta^*(\pi) \leq \delta\}$, with $F_T(0) = \tau^*$, $F_T(\infty) = \tau_{\text{free}}$, and $\Phi_T \leq F_T$.*

Positions enter only through $\tau(\pi)$ (Proposition 2) and merge levels only through $\delta^*(\pi)$ (Proposition 8). The barrier filling prescribes values only between anchors, which remain separated by the gap g , so the position-optimal layout and the optimal filling can be combined. We evaluate F_T by enumerating all $n!$ orders with both closed forms, which covers every frame in our data ($n \leq 11$).

6 CERTIFICATE-DRIVEN MERGE TREE MAPS

The theory suggests a method that keeps the merge tree where this is cheap and relaxes it only where the certificate shows a conflict. Algorithm 2 summarizes it; a worked pipeline example is in the supplement.

Reference. Without per-dataset tuning, the reference direction d is the principal direction of tracked extremum displacements, and $q_i = \langle R_i, d \rangle$ is scaled to the axis. The choice is invariant to the orientation of the data axes and captures the dominant motion (0.71 of the ERA5 2013/14 motion, compared with 0.61 for longitude; in 1999/2000 it nearly coincides with longitude, 0.89). A single direction cannot encode motion orthogonal to it; we return to this in Sec. 8.

Gate. For the small trees in our full-layout experiments, we compute the continuous τ^* and τ_{free} by order enumeration and relax only if $H > \theta = 2\%$ of the axis. This is an operating threshold, not a perceptually validated limit.

Threshold and prune. In a frame with $H > \theta$ we flatten all nodes with $\gamma_v \leq \kappa$, then undo flattenings, strongest first, while keeping a rounded display-model score $\hat{\tau}$ within half a pixel of its value with all admissible nodes flattened. The comparison uses this fixed initial value, so tolerance cannot accumulate across removals. Widths and gap are rounded to a 4096-pixel grid for this heuristic

Dataset	steps	Δt	leaves	tracks	tree / range
ERA5 1999/2000	118	12 h	1–8	135	join / 95.1 hPa
ERA5 2013/14	124	12 h	1–7	147	join / 112.1 hPa
Wildfire 2019	30	2 d	6–11	76	split / 4.2
Ring	40	–	1–7	19	split / 12.7
Gaussians	64	–	3–4	4	split / 1.5

Table 1: Datasets. Leaves: merge-tree leaves per time step; range: value range of the field (fire: $\log(1 + \text{FRP})$).

score; the final positional budget uses the exact continuous optimum. Flattening jointly and pruning afterwards handles plateaus on which only a set of flattenings lowers τ^* ; a greedy one-node-at-a-time policy stalls there (Sec. 7.4). The budget κ selects nodes by merge gap; the resulting topological cost is computed exactly by Proposition 8. The policy has no approximation guarantee, and Sec. 7.4 compares it with the exact frontier.

Layout. Within the budget $B_t = \tau^*(T') + \beta$ of the possibly relaxed tree T' , we choose the order and positions that minimize a weighted sum of distance stress, reference error, eccentricity, and a motion residual that asks each tracked anchor to move like its reference since the previous frame, subject to (D), (C), (E) and $|u_i - q_i| \leq B_t$. This is one quadratic program per candidate order; candidate orders are those with $\tau(\pi) \leq B_t$. The weights of reference error, distance stress, motion residual, and eccentricity are fixed at 4, 1, 0.5, and 0.1, each normalized by its number of terms.

Rendering and disclosure. Columns are filled along leaf arcs and LCA paths, and relaxed columns are then corrected with the barrier filling, so the displayed topological distortion is exactly $\delta^*(\pi)$. The disclosure answers three questions a reader has. *Can I read positions here?* A strip above the map shows, per time step, the certificate split into space and hierarchy cost, with θ as a reference line (Fig. 1b). *Which features are affected?* Per-feature flags mark features whose deviation exceeds θ , and in conflict steps the best witness triple is drawn on the map as a bar joining the two leaves of one subtree and a ring on the leaf that lies between them in the reference (Fig. 6a). *What did the relaxation give up?* Relaxed steps are marked, and a second strip shows the exact topological cost δ^* in data units (Fig. 1c). We draw strips rather than encoding errors in the map itself, because the map already uses colour for the field and position for the reference; full per-feature error ticks were illegible (Sec. 8).

Constants. All constants are relative to the canvas: the width scale maps the domain area to half the axis, $g = L/240$, $\beta = L/120$, $\rho = 0.5$, and $\theta = 2\%$ of the axis. On a 400-pixel map, θ is 8 pixels; we use it as an operational tolerance without claiming a perceptual detection threshold. The same values are used for all datasets; Sec. 7.2 reports $\theta = 1\%$ and 5% . The merge-gap threshold κ is not a constant but the user’s choice of operating point on the position–topology trade-off; it bounds the gaps of flattened merges, not the resulting distortion δ^* , which the strips report; we report $\kappa \in \{2\%, 20\%, \infty\}$ of the value range and use 20% in figures.

7 EVALUATION

We ask four questions. **RQ1:** how common are hierarchy conflicts in real data? **RQ2:** what do they do to motion readings? **RQ3:** what does positional fidelity cost in topology, and how close is the relaxation to the exact frontier? **RQ4:** does the certificate transfer to other hierarchy-constrained displays?

7.1 Data, baselines, and checks

Datasets (Tab. 1). ERA5 mean sea-level pressure over the North Atlantic and Europe (25–80°N, 40°W–50°E) [17] for 1999–11–17 to 2000–01–14 and for December 2013 to January 2014, 12-hourly, on an equal-area 49×49 grid with 250 km smoothing; features are

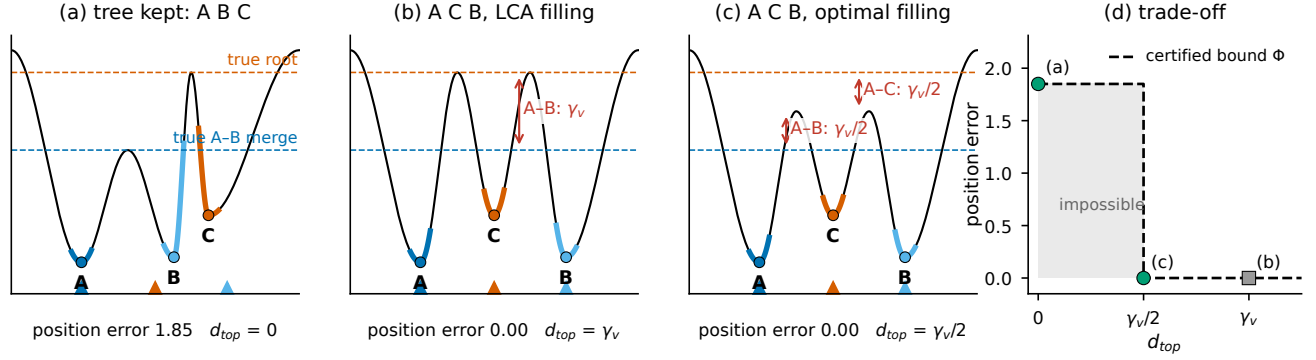


Figure 3: The price of topology on the example of Fig. 2 (columns as in Fig. 2d; triangles: references). (a) Keeping the merge tree reads every merge correctly but misplaces B and C by τ^* . (b) Placing every anchor at its reference separates A and B; with the LCA-path filling the A-B merge is read at the root, an error of γ_v . (c) The barrier filling of the same order splits the error: $d_{top} = \gamma_v/2$, the lower bound of Theorem 7. (d) The resulting trade-off: no 1-D map lies in the grey region below the certified bound Φ ; (a) and (c) lie on it, (b) does not.

Algorithm 2 Certificate-driven merge tree map. Constants: $\theta = 2\%$ of the axis; canvas-relative g, β, ρ ; budget κ .

Require: merge trees T_i with leaf extrema R_i and supports A_i ; feature tracks

- 1: $d \leftarrow$ principal direction of tracked extremum displacements;
- 2: $q_i \leftarrow \langle R_i, d \rangle$ scaled to the axis
- 3: **for** $t = 1, \dots, T$ **do**
- 4: $\tau^* \leftarrow \min_{\pi \in \Pi(T_i)} \tau(\pi)$; $\tau_{free} \leftarrow \min_{\pi \in S_n} \tau(\pi)$; $H_t \leftarrow \tau^* - \tau_{free}$
- 5: $T' \leftarrow T_i$
- 6: **if** $H_t > \theta$ **then** $\triangleright$ threshold and prune
- 7: $W \leftarrow \{v : \gamma_v \leq \kappa\}$; $b_0 \leftarrow \widehat{\tau}(\text{flat}(T_i, W))$
- 8: **for** $v \in W$ by decreasing γ_v **do**
- 9: **if** $\widehat{\tau}(\text{flat}(T_i, W \setminus \{v\})) \leq b_0 + \text{half a pixel}$ **then**
- 10: $W \leftarrow W \setminus \{v\}$
- 11: $T' \leftarrow \text{flat}(T_i, W)$
- 12: $B_t \leftarrow \tau^*(T') + \beta$ $\triangleright$ error budget
- 13: $(\pi, z, u) \leftarrow$ over $\pi \in \Pi(T')$ with $\tau(\pi) \leq B_t$, minimize distance stress, reference error, eccentricity, and motion residual w.r.t. $t - 1$ subject to (D)(C)(E) and $|u_i - q_i| \leq B_t$ $\triangleright$ one QP per order
- 14: fill column t along leaf arcs and LCA paths; clip barriers to $C_k + \delta^*(\pi)$ (Prop. 8)
- 15: strips: $\tau^*, H_t, \max_i |u_i - q_i|, \delta^*(\pi)$

pressure lows (join tree). Cropping a field to a rectangle creates minima on its boundary that are only the flanks of lows outside it: on a narrower 30–75°N, 30°W–40°E window, 62% of the leaves lie on the boundary and most conflict witnesses involve them. We therefore smooth on a grid enlarged by five cells on each side (three to the north), keep as leaves only the minima of the enlarged field that lie inside the window, and raise every other basin to its spill level (minima imposition); every leaf is then a genuine low, and 6–7% of the leaves lie within one cell of the boundary. The window is the largest that our download (20–90°N, 50°W–60°E) allows with this buffer. Australian fire radiative power at 10 km resolution [15] for November–December 2019, aggregated to two-day frames on a 64×64 grid of $\log(1 + \text{FRP})$ with a weak background term that removes zero plateaus, and with the smallest smoothing that yields at most 12 maxima per frame, which keeps the exact frontier enumerable; features are fire hot spots (split tree). Ring [24, 15] and a replica of ST-MTM’s three-Gaussian scene are synthetic controls.

Methods. Baselines are TMTM; ST-MTM, which has no public

code and was reimplemented from its description [8], with its published parameters for Ring and adapted defaults elsewhere; and a reference-anchored layout along a fixed longitude (fixed-X). Ours is evaluated with the hierarchy kept (A, the reference-anchored layout along the automatic direction) and relaxed at three merge-gap thresholds (R, Algorithm 2, $\kappa \in \{2\%, 20\%, \infty\}$ of the value range). A sorted, packed baseline uses the order of reference positions with the same widths, gaps, canvas, and secondary layout objective as R, but without the tree-order constraint; we compute its attainable topological cost under optimal filling. An exact-projection baseline places every anchor exactly at its reference, without packing or hierarchy constraints, and isolates projection loss. All methods share merge trees, supports, and feature correspondences.

Checks. Every layout is checked for legality, widths, gaps, canvas, eccentricity, and budget, and every unrelaxed frame for exact equality of the rendered merge tree. Theorem 7 and Corollary 9 are checked on every rendered frame; the rendered barrier filling attains δ^* within $1.2 \cdot 10^{-5}$ of the value range (the strict-extremum clearance); $F_T(0) = \tau^*$ holds exactly; and no hierarchy-preserving method falls below the width-free certificate.

7.2 RQ1: How common are hierarchy conflicts?

Under the automatic direction, $H > \theta$ in 34%, 24%, and 63% of the time steps of ERA5 1999/2000, ERA5 2013/14, and wildfire, with maxima of 25%, 16%, and 19% of the axis and medians of 12%, 7%, and 7% among conflict frames (Fig. 4). The space cost never exceeds 5%: some order could place the features almost exactly, and it is the hierarchy that prevents it. Conflicts are not an artefact of the threshold: for $\theta = 1\%$ and 5% the shares are 25–67% and 18–57%. Nor are they an artefact of the direction: across eight fixed directions the share ranges from 11% to 77%. And they are not an artefact of our width model: the width-free certificate exceeds θ in 31%, 22%, 63%, 50%, and 69% of the frames of the five datasets. τ_{free} is exact: a reference-sorted order is accepted only when it attains a proved lower bound; otherwise all orders are enumerated.

Prevalence depends on extraction. A conflict needs at least three leaves; frames average 3.4 leaves on ERA5 and 7.9 on fire. At a fixed 150 km ERA5 smoothing, chosen independently of enumeration limits, a certificate-only lower bound proves conflicts in at least 71% and 51% of the two winters (maximum 12 and 14 leaves). At 250/350 km, exact shares are 34/17% and 24/14%, respectively; for fire leaf caps of 8/10/12 they are 57/73/63%. Boundary minima also inflate prevalence: on the narrower window with boundary minima retained, the ERA5 shares are 58% and 60%. These are protocol-dependent rates, not general incidence estimates.

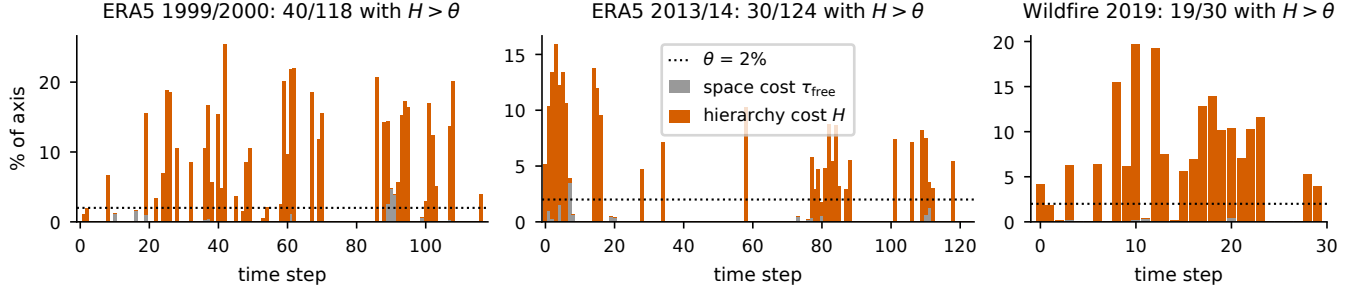


Figure 4: Per time step, space cost τ_{free} (grey) and hierarchy cost $H = \tau^* - \tau_{\text{free}}$ (orange) in % of the axis, on the three real datasets; dotted: θ . Positional distortion under the hierarchy is almost entirely hierarchy cost.

Method	ERA5	ERA5-14	Fire	Ring
TMTM	26.9 / 47.1	29.9 / 47.7	21.4 / 39.3	22.6 / 64.5
ST-MTM	21.2 / 41.3	23.4 / 53.3	21.4 / 35.7	7.3 / 34.7
fixed-X	18.3 / 24.0	19.6 / 37.4	17.9 / 36.6	23.4 / 37.1
A (tree kept)	19.2 / 23.1	15.9 / 29.9	16.1 / 32.1	24.2 / 31.5
R, $\kappa = 2\%$	17.3 / 24.0	18.7 / 29.9	12.5 / 42.9	20.2 / 22.6
R, $\kappa = 20\%$	13.5 / 18.3	13.1 / 23.4	6.2 / 31.2	10.5 / 12.9
R, $\kappa = \infty$	10.6 / 13.5	13.1 / 22.4	3.6 / 33.0	10.5 / 12.9
sorted, packed	10.6 / 14.4	13.1 / 22.4	3.6 / 33.0	8.9 / 12.1
exact projection	10.6 / 13.5	14.0 / 24.3	5.4 / 26.8	9.7 / 13.7
clear cases	104	107	112	124

Table 2: Reader proxy, $k = 2$: reversal / error (reversal + miss) in % of clear cases. Topological cost of R, largest δ^* in % of the value range, for $\kappa = 2\% / 20\% / \infty$: ERA5 1.2 / 26.1 / 58.7; ERA5-14 1.2 / 19.5 / 28.6; fire 1.7 / 19.6 / 52.6; Ring 2.2 / 12.3 / 12.3.

Simplification reduces conflicts by deleting features. Persistence simplification at 1% and 2% of the value range removes 28% and 34% of the ERA5 leaf instances (30% and 39% in 2013/14; 22% and 35% for fire), and conflicts remain in 24% and 19% of the ERA5 frames (13% and 7%; 57% and 33%). Only at 5% do conflicts fall to 3–10%, after about half of the pressure minima and 60% of the hot spots are gone. Flattening keeps every feature and pays in merge levels instead.

What existing maps could avoid. Under their best monotone calibration, TMTM and ST-MTM deviate from the reference by 11–29% of the axis on average on the real datasets, of which 2–6% is unavoidable for any merge-tree-preserving order; the remainder reflects that they optimize other goals than position (Fig. 1a).

7.3 RQ2: What do conflicts do to motion readings?

A reader of a static map may judge approach or separation by a pair’s vertical distance change. We simulate this geometric reading over $k = 2$ steps (24 h for ERA5, four days for fire), with a 2% change threshold relative to each method’s occupied anchor range. We use two separate truths: change in 2-D extremum distance beyond 2% of the domain diagonal (Tab. 2), and change in projected-reference distance beyond 2% of the reference-axis extent. A *reversal* reads a clear change as its opposite, a *miss* as no change; their sum is the error. Each clear pair-window is one case. Paired differences use a circular moving-block bootstrap over time with $k + 2$ -step blocks. If every anchor error is at most B_t at time t , then the displayed and projected pair-distance changes differ by at most $2(B_t + B_{t+k})$; this sufficient reading margin concerns the projection, not 2-D motion.

Existing maps and exact projection. Table 2 shows that existing maps reverse 21–30% of clear readings on real data, below but not far from the permutation null (32–39%). Even the exact-

projection baseline, which draws every feature at its reference, reverses 11–14% on pressure and 5% on fire, because a single direction cannot show motion orthogonal to it. It is a reference point, not a bound: R at $\kappa = \infty$ on fire and ERA5 2013/14 and ST-MTM on Ring fall below it. Keeping the merge tree (A) lies 9 points above it on ERA5 1999/2000, consistent with the hierarchy cost of RQ1, and 2 points above it on 2013/14, where conflicts are rarer.

Projection versus layout loss. Against projected truth, R at $\kappa = 20\%$ reverses 5.8%, 0%, 2.8%, and 0% of clear cases on ERA5, ERA5-14, fire, and Ring, versus 13.5%, 13.1%, 6.2%, and 10.5% against 2-D truth. The projected clear-case counts are 120, 107, 107, and 140, respectively (the 2-D counts are in Tab. 2); denominators differ because the two truths define different clear events. Exact projection has zero projected error by construction. R still misses 5.8%, 4.7%, 15.0%, and 0.7% of projected cases because packing and the operating budget remain. The contrast shows why the reference task and 2-D task must be reported separately; their rates cannot be subtracted to estimate a projection effect.

Simple sorting baseline. The packed reference sort is competitive when topology is unrestricted: its 2-D reversal/error is 10.6/14.4%, 13.1/22.4%, 3.6/33.0%, and 8.9/12.1% (Tab. 2). Its largest attainable merge-level distortion is 58.7%, 28.6%, 52.6%, and 12.3% of each value range, compared with 26.1%, 19.5%, 19.6%, and 12.3% for R at $\kappa = 20\%$. On Ring it also has a lower reversal rate than R at the same largest cost. R’s contribution is therefore a disclosed, selectable position–topology trade-off, not dominance of simple sorting on the proxy.

Relaxation as a trade-off. Benefit and cost both depend on the merge-gap threshold κ . At $\kappa = 2\%$ the largest merge-level error stays below 2% of the value range, and reversals on pressure stay within 3 points of A. At $\kappa = 20\%$, reversals drop to 13–14% on pressure and 6% on fire, below ST-MTM by 7.7, 10.3, and 15.2 points (95% intervals $[-17.0, +2.0]$, $[-19.6, -1.0]$, $[-26.2, -3.9]$; the first is not significant, and fire has 8 blocks and is descriptive). The combined error drops by 23.1 and 29.9 points on pressure ($[-34.0, -12.7]$, $[-39.8, -20.0]$), because the relaxed map also misses fewer changes; on fire it drops by 4.5 points, which is not significant, as reversals turn into misses. The price is a largest merge-level error of 20–26% of the value range, with a mean over frames of 1.5–4%. At $\kappa = \infty$, R reaches the exact-projection baseline at a largest merge-level error of 29–59%. On Ring, where ST-MTM uses its published parameters, it reverses less (7.3% vs 10.5%, not significant) but misses far more (error 35% vs 13%).

Robustness. At $\kappa = 20\%$, using fixed x instead of the fitted direction gives reversal rates of 12–13% on the real datasets. Across three reader thresholds and four time windows, R reverses less often than ST-MTM in 16 of 18 settings, significantly in 9 (exploratory, without multiplicity correction); a 5% threshold ties, and the 4-step ERA5-14 window is an exception. A grid covering the four published ST-MTM presets shows that settings with few reversals

often freeze the layout (up to 95% misses); the best combined error remains above R's on all four datasets. Extraction matters: at 150 km smoothing ST-MTM's optimizer fails on ERA5 1999/2000, and with a coarser fire extraction (leaf cap 8) relaxation does not lower reversals. The proxy assumes exact feature identities and anchors; it does not measure human performance.

Does the certificate locate errors? A frame-level flag ($H > \theta$) marks most clear cases and does not predict where TMTM or ST-MTM reverse. A per-feature flag is more informative. We mark a pair when either feature deviates by more than θ under the best monotone calibration of its map (Proposition 6), so that the flag applies to any map. It catches 75–89% of the reversals of A on real data, and flagged cases are 1.8–3.2 times as likely to be reversed. It is not selective, marking 51–71% of clear cases, and unflagged cases still reverse in 6–13%. For R at $\kappa = 20\%$ few cases remain flagged (2–32%), and the flag catches 0–57% of the remaining reversals. The flag is a diagnostic, not a guarantee.

7.4 RQ3: What does fidelity cost, and is the relaxation optimal?

Figure 5 shows, per dataset, the exact frontier F , the certified bound Φ , and our layouts. Theorem 7 holds in all 656 rendered frames with broken contiguity, and no layout lies below Φ or F .

Cost. At $\kappa = 2\%$, relaxation lowers the mean reference error by 9–17% on pressure at 1.1–1.3 hPa of merge-level error with the optimal filling (1.6–2.2 hPa with the LCA filling), but by 45% on fire. At $\kappa = 20\%$ it lowers the error by 48–62% on pressure at 22–25 hPa. The cost of positional fidelity depends on the merge structure: fire conflicts stem from weak merges between neighbouring hot spots, pressure conflicts from strong saddles and shallow lows.

Optimality on relaxed frames. The selected order's $(\tau(\pi), \delta^*(\pi))$ lies within half a pixel of the exact frontier in 44/44, 99/102, and 106/110 relaxed frames at $\kappa = 2\%, 20\%, \infty$ (four datasets), never more than 0.7% of the axis above it. The threshold-and-prune policy's τ -optimal orders are on the frontier in every relaxed frame. Greedy one-node flattening stalls on plateaus, resolving only 88/110 conflicts at $\kappa = \infty$; flattening without pruning changes the tree in 73 rather than 44 frames at $\kappa = 2\%$. The aggregate points lie above F because κ thresholds merge gaps, whereas actual δ^* also includes leaf heights.

Tightness of the certified bound. $F - \Phi$ averages 0.3–1.2% of the axis over all breakpoints of Φ and is within half a pixel at 29–67% of them; it exceeds 10% in a few pressure and fire frames (at most 22%). Φ ignores the leaf-height term of Proposition 8, which can only widen this gap. The optimal filling halves d_{top} where only merge gaps matter (Ring, Gaussians) but saves little on pressure at large budgets, again because of shallow lows.

Hidden costs. Relaxation could harm properties that the reader proxy does not measure. Table 3 reports method-agnostic measures. On fire, tracked pairs swap vertical order in 3% of consecutive steps on R, against 8% and 33% on ST-MTM and TMTM; on ERA5, R (7%) lies between ST-MTM (5%) and TMTM (9%), and on Ring ST-MTM swaps fewer. Displacement-based stability is mixed: on pressure data R's anchors move more per step than ST-MTM's, as they follow the lows. R's 2-D distance preservation is worse than that of ST-MTM on pressure data (nearest-neighbour preservation 0.70 vs 0.88 on ERA5) and similar to that of exact projection (0.66), so much of this cost comes from reading a single direction; ST-MTM is the better choice when 2-D neighbourhoods matter more than positions along a direction. At $\kappa = 20\%$ the barrier filling changes 2–6% of the pixels, all in relaxed columns; unrelaxed columns are untouched.

7.5 Case study: reading a winter with the certificate

Figure 6 shows how an analyst would use the certificate on ten days of the ERA5 winter. The hierarchy strip of the merge-tree-

Method	flips	jitter	stress	NN
TMTM	.092 / .334	.046 / .102	.317 / .458	.746 / .639
ST-MTM	.050 / .076	.027 / .053	.194 / .256	.877 / .753
A	.110 / .094	.101 / .054	.327 / .261	.731 / .658
R, $\kappa = 20\%$	.073 / .026	.089 / .022	.308 / .221	.700 / .630
R, $\kappa = \infty$	.083 / .010	.086 / .019	.296 / .219	.682 / .600
exact projection	.078 / .005	.086 / .017	.310 / .227	.660 / .589

Table 3: Hidden costs on ERA5 / fire: share of tracked pairs that swap vertical order between consecutive steps (flips), mean anchor displacement per step relative to the occupied range (jitter), Kruskal stress-1 and nearest-neighbour preservation (NN) of 1-D against 2-D extremum distances. Lower is better for the first three, higher for NN.

preserving map (a) exceeds θ in eight of the twenty steps, up to 25% of the axis, mostly between 1999-12-06 and 1999-12-11: on these days positional readings require caution. The witness triples explain why. At step 48 (1999-12-11 00 UTC), a deep low of approximately 985 hPa west of Scotland and one of 988 hPa over Scandinavia merge at 994.1 hPa, before either merges with a shallower 1007 hPa low over the Adriatic that lies between them along the reference direction (b). Any map that keeps the merge tree must place the 1007 hPa low outside the interval of the other two (c); this witness alone forces 7.8% of the axis, most of the certificate (8.4%). If the tracks matter, the relaxed map (d) removes the conflict, with an error of 0.8% of the axis at step 48. The topological price is explicit: even with the optimal filling, the west-Scotland/Scandinavia merge reads at 1006.7 rather than 994.1 hPa, an error of 12.6 hPa. An analyst studying the depth of that shared trough should keep the tree-preserving view.

7.6 RQ4: Does the certificate transfer?

Dense pixel displays often order rows by a dendrogram. We ordered 1024 active 10-km fire cells, one pixel row each, by an agglomerative clustering of their fire time series, the hierarchy behind hierarchical orderings of pixel displays [15]. The reference is each cell's position along the principal geographic axis, scaled linearly to the display height. The certificate shows that any order consistent with this dendrogram must displace some cell by at least 67% of the display height, whereas the geographic order itself stays within 21%. With a rank-based reference the bound is 78–82%. The hierarchy cost is 47–48% for average, complete, and Ward linkage, versus 1.4% for a spatial Ward dendrogram and 32.5% for a spatial single-linkage dendrogram. With optimal leaf ordering, cells sit up to 88% (median 14%) away from geography, and 8% of the rows exceed the certificate (supplementary dendrogram figure). The certificate thus answers a design question before anyone reads geography off a clustered display: here, “row position approximates location” is untenable, and a linked map view or a spatially constrained hierarchy is needed. The computation takes 0.1 s.

Runtime. On recorded low-degree trees, the certificate takes about 2 ms per small frame at 1024 pixels and 3.6 s for 4096 leaves at 16384 pixels. In a fixed 12-leaf, 96-pixel stress test, flattening the root from degree 2 to 12 raises subset states from 3 to 4095 and DP time from 0.001 to 0.135 s. Frontiers enumerate $n!$ orders; layout selection also enumerates orders and solves QPs. Large-scale evidence applies to diagnosis only.

8 DISCUSSION AND LIMITATIONS

Implications. A hierarchy-preserving map should disclose certified deviation per time step and feature. If position matters more, the trade-off curve and actual topological cost should guide relaxation. Reporting achieved error beside the certificate and exact projection separates layout, hierarchy, and projection loss.

Limits of the theory. One reference coordinate cannot encode orthogonal motion; radial or path references need separate evalua-

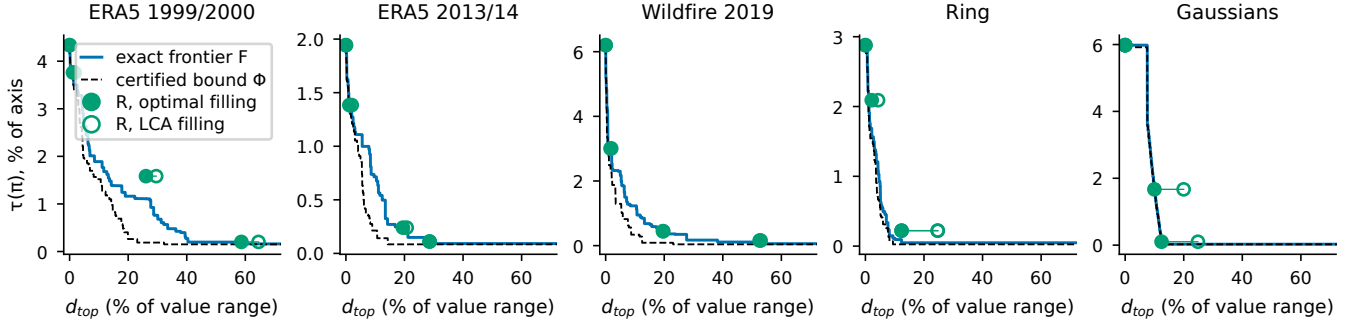


Figure 5: Exact frontier F (blue), certified bound Φ (dashed), and R at four κ values (filled: optimal filling; open: LCA filling). x : largest topological error over frames; y : mean fixed-order optimum $\tau(\pi)$, excluding layout slack β .

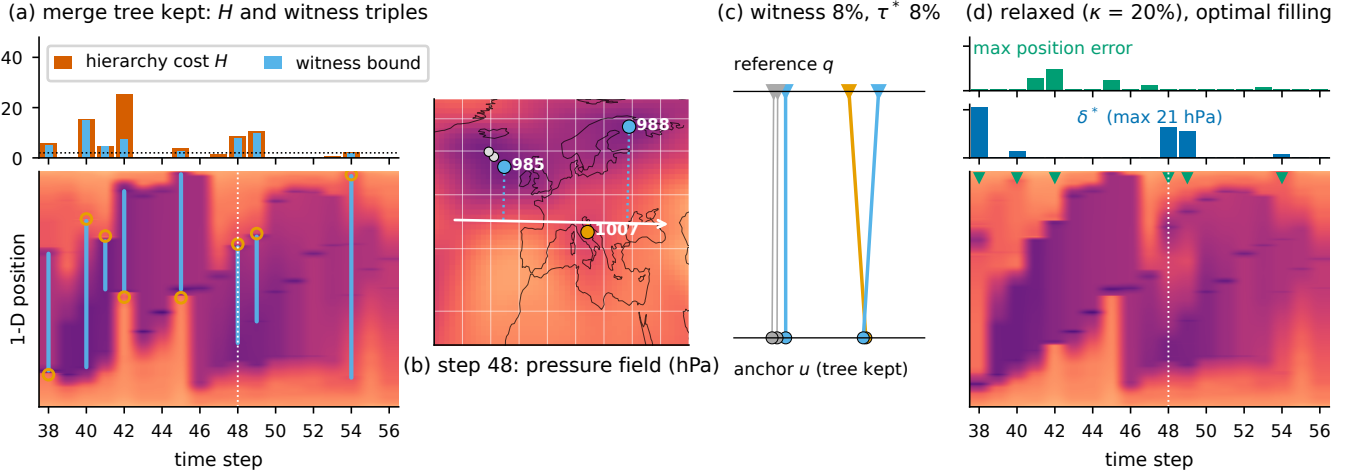


Figure 6: ERA5 winter case. (a) Tree-preserving map: blue bars and orange rings mark witness triples in conflict steps. (b) Step 48 field and projected extrema. (c) References versus tree-preserving anchors: the orange low cannot lie between the blue lows. (d) Relaxed map ($\kappa = 20\%$) with position-error and topological-cost strips.

tion. The optimal filling and frontier assume true extrema values. The certificate DP is pseudo-polynomial in resolution and exponential in node degree; layout selection and the exact frontier enumerate orders. Complexity for unbounded degree remains open. Φ ignores leaf heights and becomes expensive after flattening. Relaxation has no approximation guarantee, and κ thresholds merge gaps rather than actual topological cost.

Limits of the evaluation. The geometric proxy does not establish human performance. The direction is fitted on the evaluated tracks, though the fixed-direction control retains the reversal gain. ST-MTM was reimplemented from its description; we could not compare with its original code. ERA5 comes from a public mirror and is not byte-identical to the original files; window and boundary-minimum choices affect conflict rates (Sec. 7.2). At 150 km smoothing we report certificate lower bounds only, not full relaxation or motion results. The fire background term was not varied.

Negative results. Radial centres followed tracking artefacts, stability-priority relaxation did not generalize, and per-feature error ticks were illegible at map scale; witness triples are shown only in conflict steps.

9 CONCLUSION

The positional certificate quantifies unavoidable reference error and identifies conflicts. The optimal filling and exact small-tree frontier

quantify what changes when contiguity is relaxed. In our selected protocols, 24–63% of frames conflict; reducing geometric motion-proxy error can incur substantial merge-level distortion. The appropriate trade-off depends on the task, reference, and extraction choices.

FIGURE CREDITS

Figures: authors. Fire data: Franke et al. [15] (CC BY 4.0); ERA5: Copernicus [17]; coastlines: Natural Earth (public domain).

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